

1.

Let the conditional densities for a two-category one-dimensional problem be given by the Cauchy distribution

$$p(x|\omega_i) = \frac{1}{\pi b} \cdot \frac{1}{1 + \left(\frac{x-a_i}{b}\right)^2}$$

with $i = 1, 2$.

(a)

If $P(\omega_1) = P(\omega_2)$, show that $P(\omega_1|x) = P(\omega_2|x)$ if $x = (1/2)(a_1 + a_2)$. Sketch $P(\omega_1|x)$ for the case $a_1 = 3, a_2 = 2, b = 5$. How does $P(\omega_1|x)$ behave as $x \rightarrow -\infty$? as $x \rightarrow \infty$?

(b)

Using the conditional densities in part a, and assuming equal *a priori* probabilities, show that the minimum probability of error is given by

$$P(\text{error}) = \frac{1}{2} - \frac{1}{\pi} \tan^{-1} \left| \frac{a_2 - a_1}{2b} \right|.$$

Sketch this as a function of $|(a_2 - a_1)/b|$.

2.

The Poisson distribution for a discrete $k, k = 0, 1, 2, \dots$ and real parameter λ is

$$P(k|\lambda) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

(a)

Find the mean of k .

By definition, we have that

$$\begin{aligned} E[k] &= \sum_{k=0}^{\infty} k \cdot P(k|\lambda) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} \lambda^k e^{-\lambda} \frac{k}{k!} \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \lambda^k \frac{1}{(k-1)!} \end{aligned}$$

Simplifying by substituting $j = k - 1$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \lambda^{j+1} \frac{1}{j!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= \lambda e^{-\lambda} \cdot e^{\lambda} \\ &\boxed{E[k] = \lambda} \end{aligned}$$

(b)

Find the variance of k .

We need to essentially find $E[k^2]$ here. We can quickly ascertain that directly substituting k^2 will make this hard to simplify and cancel out the k terms in both the numerator and factorial denominator, so we can instead solve for $E[k(k-1)]$:

$$E[k(k-1)] = \sum_{k=0}^{\infty} k(k-1) \cdot e^{-\lambda} \frac{\lambda^k}{k!}$$

$$\begin{aligned}
&= \sum_{k=0}^{\infty} e^{-\lambda} \frac{(k-1) \cdot \lambda^k}{(k-1)!} \\
&= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-2)!} \\
&= 0 + 0 + e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}
\end{aligned}$$

Simplifying by substituting $j = k - 2$:

$$\begin{aligned}
&= e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+2}}{j!} \\
&= \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\
E[k(k-1)] &= \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2 \\
E[k^2 - k] &= \lambda^2 \\
E[k^2] - E[k] &= \lambda^2 \\
E[k^2] &= \lambda^2 + E[k] = \lambda^2 + \lambda
\end{aligned}$$

We can now solve for variance:

$$\begin{aligned}
\text{Var}(k) &= E[k^2] - E[k]^2 \\
&= \lambda^2 + \lambda - \lambda^2 \\
\boxed{\text{Var}(k) &= \lambda}
\end{aligned}$$

(c)

Find the mode of k .

The mode is the value of k with the largest probability of occurring. Since k is an integer value, it must be an integer value. We can determine this by observing the nature in which the sequence continues element by element:

$$P(k|\lambda) =$$

$$P(k+1|\lambda) = e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}$$

We see that

$$\begin{aligned} \frac{P(k|\lambda)}{P(k+1|\lambda)} &= \frac{e^{-\lambda} \frac{\lambda^k}{k!}}{e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}} \\ &= \frac{\lambda^k \cdot (k+1)!}{\lambda^{k+1} \cdot k!} \\ &= \frac{k+1}{\lambda} \end{aligned}$$

So essentially, with the way this is set up, we want to find when the current probability is bigger than the next probability, since we know the Poisson distribution is parabolically shaped a maximum point after which the probability begins decreasing. We have a few different cases:

- If $\frac{k+1}{\lambda} > 1$, then probabilities are going down.
- If $\frac{k+1}{\lambda} < 1$, the probabilities are going up.
- If $\frac{k+1}{\lambda} = 1$, then $k = \lambda - 1$.

From looking at this, we can see that the highest probability or mode happens around $k = \lambda - 1$. However, since it is possible for $k = \lambda - 1$, in this case in which λ is an integer, we need to differentiate two cases, since both $\boxed{\lambda - 1}$ and $\boxed{\lambda}$ have equal probability and are both integers. In the case in which λ is not an integer, then since k must be an integer, we just take $\boxed{\lambda \text{ floored to the nearest integer}}$.

(d)

Assume two categories C_1 and C_2 , equally probably *a priori*, distributed with Poisson distributions and $\lambda_1 > \lambda_2$. What is the Bayes classification decision?

We have $P(C_1) = P(C_2) = 0.5$, and $P(x|C_1) = e^{-\lambda_1} \frac{\lambda_1^x}{x!}$ and $P(x|C_2) = e^{-\lambda_2} \frac{\lambda_2^x}{x!}$. By definition, we decide the category based on the probability comparison, since the priors are equal, i.e. we would choose C_1 if $P(C_1) > P(C_2)$, but as they are equal,

we see if $P(x|C_1) > P(x|C_2)$:

$$\begin{aligned}
e^{-\lambda_1} \frac{\lambda_1^x}{x!} &> e^{-\lambda_2} \frac{\lambda_2^x}{x!} \\
e^{-\lambda_1} \lambda_1^x &> e^{-\lambda_2} \lambda_2^x \\
-\lambda_1 + x \ln(\lambda_1) &> -\lambda_2 + x \ln(\lambda_2) \\
x \ln(\lambda_1) - x \ln(\lambda_2) &> -\lambda_2 + \lambda_1 \\
x(\ln(\lambda_1) - \ln(\lambda_2)) &> \lambda_1 - \lambda_2 \\
x &> \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}
\end{aligned}$$

Therefore, the decision rule is as follows:

Choose C_1 if $x > \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$; Otherwise, choose C_2 .

(e)

What is the Bayes error rate?

Logically, the Bayes error rate is the probability of making an incorrect classification; that is, either we choose C_2 when C_1 was the right category, or we choose C_1 when C_2 was the right category. Using our priors with $P(C_1) = P(C_2) = 0.5$, and letting the threshold of classification to be $y = \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$, with the understanding that we classify into the categories based on being above or below this threshold, we can express $P(\text{error})$ to be

$$P(\text{error}) = P(C_1) \cdot \sum_{k=0}^{\lfloor y \rfloor} P(k|C_1) + P(C_2) \cdot \sum_{k=\lceil y \rceil}^{\infty} P(k|C_2)$$

$$P(\text{error}) = \frac{1}{2} \cdot \sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \frac{1}{2} \cdot \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}$$

$P(\text{error}) = \frac{\sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}}{2}$
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3.

(a)

(b)

4.